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Paper lid for food container and method of manufacturing the same

Abstract

Disclosed is a method of manufacturing a paper lid for a food container, the method including a blanking step of cutting out lid molding paper having radial wrinkles at the edge thereof from base paper having upper and lower surfaces, each of which is treated with a synthetic resin coating, a primary molding step of molding the paper lid including a cover plate and a skirt wall bent downwards along the edge of the cover plate, and a secondary molding step of forming a guide part obtained by bending a large diameter part of the skirt wall so as to be inclined outwards, the guide part guiding coupling with the upper portion of the food container, and molding a curling tight-contact part configured to tightly surround the outside of a reinforcement frame of the food container by curling the upper end of the guide part in the inward direction.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a paper lid for a food container and a method of manufacturing the same, and more particularly to a paper lid for a food container and a method of manufacturing the same, configured to increase strength of the food container by curling the edge of the paper lid using an eco-friendly paper material and to be reliably applied to the food container.

BACKGROUND ART

(2) In general, a food container is applied to various fields. For example, the food container is used to contain food (instant food) such as cup noodles and a beverage such as water or take-out coffee. Recently, a disposable food container made of a paper material has been used to solve environmental problems.

(3) In addition, a lid used for the disposable paper container is also made of a paper material in consideration of environmental problems. In Korea Patent Registration No. 10-0568560 (entitled "paper lid with improved compatibility"), disclosed is a paper lid structure manufactured using a paper material and configured not only to maintain a stable coupling state with a cup container but also to improve product design.

(4) As shown in FIG. **1**, the paper lid with improved compatibility disclosed in Korea Patent Registration No. 10-0568560 has a lid body structure including a flange part **20** bent downwards and formed to extend in the vertical direction along the peripheral edge of a main body part **10** forming a circular planar shape, wrinkled grooves **21** formed along the flange part **20** at a predetermined distance, and a detachable groove **11** formed along the inner corner portion where the flange part **20** and the main body part **10** meet. Therefore, in the process of coupling the lid body to a cup container **30**, even if the diameter of a curling part **31** of the cup container **30** is slightly different depending on the cup container **30**, the size of the flange part **20** is changed depending on the diameter of the curling part **31**, thereby having an effect of improving compatibility.

(5) However, in the structure of the "paper lid with improved compatibility" as described above, since the size of the flange part **20** is changed, there is a disadvantage in that the lid body may be easily separated from the curling part **31** of the cup container **30** and the contents of the cup container **30** may be poured out. Particularly, since only the upper end of the curling part **31** of the cup container **30** is accommodated in the detachable groove **11** of the lid body to maintain a close contact state therebetween, there is a limit to increasing airtightness. Further, in consideration of compatibility, since the detachable groove **11** is formed along the inner corner portion where the flange part **20** and the main body part **10** meet, the lid body may be easily detached from the cup container **30**.

DISCLOSURE

Technical Problem

(6) Therefore, the present invention has been made in view of the above problems, and it is an object of the present invention to manufacture a paper lid capable of covering a food container with paper having a predetermined thickness and to provide "the paper lid for the food container and a method of manufacturing the same", configured to more stably maintain a coupling state between the paper lid and a curling part (reinforcement frame) of the food container by improving an edge structure of the paper lid and to significantly improve strength of the food container.

Technical Solution

(7) In accordance with an aspect of the present invention, the above and other objects can be accomplished by the provision of a method of manufacturing a paper lid for a food container, the paper lid covering an upper portion of the food container having a reinforcement frame formed at an uppermost portion of a wall surface and curled outwards so as to have a circular cross-sectional shape, the method including a blanking step of cutting out a lid molding paper having a size capable of molding the paper lid from base paper having upper and lower surfaces, each of which is

treated with a synthetic resin coating having a thickness of 10 to 35 microns, and forming wrinkles radially on an edge of the lid molding paper so as to facilitate molding of the paper lid, a primary molding step of supplying the lid molding paper made in the blanking step to a primary molding device, compressing, while applying heat to the lid molding paper at temperature of 100° C. to 140° C., the lid molding paper to mold the paper lid including a cover plate configured to cover the open upper portion of the food container while being spaced apart from the open upper portion and a skirt wall bent downwards along an edge of the cover plate and coupled to the food container so as to surround an outside of the reinforcement frame of the food container, and bending the skirt wall so as to form a small diameter part connected to the edge of the cover plate, the small diameter part having a lower end in contact with an upper end of the food container, and a large diameter part formed to expand in diameter while forming a round-shaped stepped part configured to allow the lower end of the small part to contact an upper end of the reinforcement frame of the food container, the large diameter part surrounding an outside of the upper end of the food container while being spaced apart from the outside thereof, and a secondary molding step of moving the paper lid made in the primary molding step to a secondary molding device, forming, while applying heat to the paper lid at temperature of 100° C. to 140° C., a guide part obtained by bending the large diameter part of the skirt wall so as to be inclined outwards in a direction toward a lower end thereof, the guide part guiding coupling with the upper portion of the food container, and molding a curling tight-contact part configured to tightly surround the outside of the reinforcement frame of the food container by curling an upper end of the guide part in an inward direction so as to have a semicircular round cross-sectional shape while the upper end thereof is connected to the round-shaped stepped part.
(8) Further, the synthetic resin coating may be made of at least one of polyethylene (PE), polypropylene (PP), polylactic acid (PLA), polyethylene terephthalate (PET), or silicone.
(9) Additionally, a cover reinforcement part may be molded in the primary molding step, the cover reinforcement part being concavely recessed downwards at a central portion of the cover plate.
(10) In accordance with another aspect of the present invention, there is provided a paper lid for a food container manufactured by the above-described method, wherein the paper lid is manufactured by cutting out lid molding paper from base paper and allowing the lid molding paper to pass through a primary molding step and a secondary molding step.
Advantageous Effects
(11) According to a paper lid for a food container of the present invention and a method of manufacturing the same, the paper lid capable of covering the food container is formed with paper having a predetermined thickness, and a skirt wall formed on the edge of the paper lid includes a small diameter part in contact with the upper end of the food container, a guide part formed by bending a large diameter part extending from the lower end of the small diameter part so as to be inclined outwards in the direction toward the edge thereof, the guide part guiding coupling between the paper lid and the upper portion of the food container, and a curling tight-contact part configured to tightly surround the outside of a reinforcement frame of the food container by curling the end of the small diameter part and the boundary portion of the large diameter part in the inward direction so as to have a semicircular round cross-sectional shape. Therefore, there is an advantage in that, when the upper portion of the food container is covered with the paper lid, the curling tight-contact part provided in the middle of the skirt wall may stably cover the open upper portion of the food container by entirely surrounding the outside of the reinforcement frame of the food container. In addition, the guide part, formed at the edge of the skirt wall and molded to be inclined outwards, has an effect of allowing the paper lid to smoothly cover the food container downwards from the upper portion of the food container.
(12) Further, according to the method of manufacturing the paper lid for the food container of the present invention, a synthetic resin coating is made of at least one of polyethylene (PE), polypropylene (PP), polylactic acid (PLA), polyethylene terephthalate (PET), or silicone. Since the

paper lid is manufactured while applying heat thereto at the temperature of 100° C. to 130° C. in the primary and secondary molding steps, there is an advantage in that the paper lid may be molded more easily and stably.
(13) Additionally, according to the method of manufacturing the paper lid for the food container of the present invention, bending strength is increased by molding a cover reinforcement part concavely recessed downwards at the central portion of a cover plate, thereby having an effect of further improving product reliability of the paper lid.

Description

DESCRIPTION OF DRAWINGS
(1) The above and other objects, features and other advantages of the present invention will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a diagram showing a paper lid with improved compatibility disclosed in Korean Patent Registration No. 10-0568560;
(3) FIG. **2** is a perspective view showing a food container to which the present invention is applied and a paper lid for the food container according to the present invention;
(4) FIG. **3** is a cross-sectional view taken along line III-III in FIG. **2**;
(5) FIG. **4** is a process block diagram sequentially showing a method of manufacturing the paper lid for the food container according to the present invention;
(6) FIG. **5** is a diagram schematically showing a blanking step in the method of manufacturing the paper lid according to the present invention;
(7) FIG. **6** is a diagram schematically showing a primary molding step in the method of manufacturing the paper lid according to the present invention and the paper lid molded in the primary molding step;
(8) FIGS. **7** to **9** are diagrams schematically showing a secondary molding step in the method of manufacturing the paper lid according to the present invention and the paper lid molded in the secondary molding step; and
(9) FIGS. **10** and **11** are cross-sectional views sequentially showing the process of covering the upper portion of the food container with the paper lid according to the present invention.
REFERENCE NUMBER LIST
(10) **100**: food container **110**: wall surface **111**: reinforcement frame **120**: bottom **200**: paper lid **210**: cover plate **211**: cover reinforcement part **220**: skirt wall **221**: small diameter part **222**: stepped part **223**: large diameter part **224**: guide part **225**: curling tight-contact part **300**: base paper **310**: paper pulp **320**: synthetic resin coating **330**: lid molding paper **331**: wrinkle **400**: primary molding device **500**: secondary molding device S**10**: blanking step S**20**: primary molding step S**30**: secondary molding step
Best Mode
(11) Reference will now be made in detail to the preferred embodiments of the present invention, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.
(12) Hereinafter, one preferred embodiment according to the present invention will be described in detail with reference to the accompanying drawings.
(13) <Description of the configuration of a paper lid according to the present invention>
(14) First, as shown in FIGS. **2** and **3**, a food container **100** to which the present invention is applied is formed of a cylindrical wall surface **110** and a bottom **120** made of paper. The uppermost portion of the wall surface **110** has a reinforcement frame **111** curled outwards so as to have a circular cross-sectional shape. Here, the wall surface **110** of the food container **100** has a tapered

tube shape formed by rolling single-layer paper pulp so as to have a wider upper portion and a gradually narrow lower portion and overlapping the opposite ends of the paper to bond the same together. In this case, the upper portion has the reinforcement frame **111** rolled outwards and provided to increase strength of the food container.

(15) A paper lid **200** according to the present invention applied to the food container **100** as described above is molded through lid molding paper acquired from base paper by cutting out the same so as to have radial wrinkles on the edge thereof. The paper lid **200** includes a cover plate **210** configured to cover an open upper portion of the food container **100** while being spaced apart from the open upper portion, and a skirt wall **220** bent along the edge of the cover plate **210** and coupled to the food container **100** so as to surround the outside of the reinforcement frame **111** of the food container **100**. A cover reinforcement part **211** concavely recessed downwards is formed at the center of the cover plate **210** of the paper lid **200**. Further, the skirt wall **220** of the paper lid **200** includes a small diameter part **221** connected to the edge of the cover plate **210** and configured to contact the upper end of the food container **100**, a guide part **224** (large diameter part) formed to have a step at the end of the small diameter portion **221** and configured to expand in diameter to surround the outside of the uppermost end of the food container **100** while being spaced apart therefrom, and a curling tight-contact part **225** configured to tightly surround the outside of the reinforcement frame **111** of the food container **100** by curling the end of the small diameter part **221** and a boundary portion of the guide part **224** in the inward direction so as to have a round cross-sectional shape.

(16) In order to make the paper lid **200** of the food container according to the present invention, a lid molding paper having a disc shape and a size capable of molding the paper lid **200** is cut out from base paper, which is sanitary and eco-friendly single-layer paper pulp, and these pieces of lid molding papers are continuously supplied from a lid molding device to make the paper lids **200**.

(17) <Description of a method of manufacturing the paper lid of the disposable container according to the present invention>

(18) Next, a method of manufacturing the paper lid according to the present invention configured as described above will be described with reference to FIGS. **4** to **9**.

(19) As shown in FIG. **4**, the method of manufacturing the paper lid for the food container according to the present invention includes a blanking step (S**10**) of cutting out, from base paper, lid molding paper having a disc shape and a size capable of forming the paper lid **200**, a primary molding step (S**20**) of primarily molding the paper lid **200** including the cover plate **210** and the skirt wall **220** using the lid molding paper made in the blanking step (S**10**), and a secondary molding step (S**30**) of molding the curling tight-contact part **225** so as to have a round cross-sectional shape in the middle of the skirt wall **220** of the paper lid **200** made in the primary molding step (S**20**) and simultaneously molding the guide part **224**.

(20) Referring to FIG. **5**, the blanking step (S**10**) is a process of cutting out, from base paper **300**, a lid molding paper **330** having a disc shape and a size capable of molding the paper lid **200**. The base paper **300** uses a single-layer paper pulp **310** having upper and lower surfaces, each of which is treated with a synthetic resin coating **320** so as to prevent moisture from penetrating the base paper **300**. The lid molding paper **330** having the disk shape and the size capable of molding the paper lid **200** is cut out through a wooden mold from the base paper **300** (referred to as “Thomson process”). In this process, it is preferable to radially form wrinkles **331** on the edge of the lid molding paper **330** so as to facilitate lid molding. That is, when pressure is radially applied to the edge of the lid molding paper **330** to form the wrinkles **331**, the lid molding paper **330** is deformed along the wrinkles **331**, thereby easily molding the paper lid **200**. In the embodiment of the present invention, the base paper **300**, which is a raw material of the lid molding paper **330**, is the single-layer paper pulp **310**, and each of the upper and lower surfaces of the paper pulp **310** is preferably treated with the synthetic resin coating **320** with a thickness of about 10 to 35 microns (μm). Additionally, the synthetic resin coating **320** treated on the base paper **300** preferably uses an eco-

friendly synthetic resin material of a polymer material that is not absorbed into the body and is not harmful to health. For example, at least one of polyethylene (PE), polypropylene (PP), polylactic acid (PLA), polyethylene terephthalate (PET), or silicone may be adopted and used as the synthetic resin coating **320**. This synthetic resin coating **320** has a function of not only preventing moisture from penetrating the paper pulp **310** but also facilitating molding of the paper lid **200**.

(21) As shown in FIG. **6**, the primary molding step (S**20**) is a process of supplying the lid molding paper **330** made in the blanking step (S**10**) to a primary molding device **400** and molding, while applying heat to the molding paper **330**, the paper lid **200** including the cover plate **210** configured to cover an open upper portion of the food container **100** while being spaced apart from the open upper portion and the skirt wall **220** bent downwards along the edge of the cover plate **210** and coupled to the food container **100** so as to surround the outside of the reinforcement frame **111** of the food container **100**. The cover plate **210** includes the cover reinforcement part **211** corresponding to the central portion of the lid molding paper **330**, the cover reinforcement part **211** being formed at the central portion of the cover plate **210** and concavely recessed downwards. The cross-section of the cover plate **210** is changed by the cover reinforcement part **211** to increase bending strength (also known as flexural strength), thereby making a more durable paper lid **200**. In addition, in the primary molding step (S**20**), the skirt wall **220** is bent to form the small diameter part **221** connected to the edge of the cover plate **210**, the small diameter part **221** having the lower end thereof in contact with the upper end of the food container **100**, and a large diameter part **223** formed to expand in diameter while forming a round-shaped stepped part **222** so as to allow the lower end (end) of the small diameter portion **221** to contact the upper end of the reinforcement frame **111** of the food container **100**, the large diameter part **223** surrounding the outside of the uppermost end of the food container **100** while being spaced apart from the outside thereof. As described above, the wrinkles **331** are radially formed on the edge of the lid molding paper **330** in the blanking step (S**10**), thereby making it possible to easily mold the skirt wall **220** including the small diameter part **221** and the large diameter part **223** in the primary molding step (S**20**).

(22) As shown in FIGS. **7** to **9**, the secondary molding step (S**30**) is a process of moving the paper lid **200** made in the primary molding step (S**20**) to a secondary molding device **500**, forming, while applying heat to the paper lid **200**, the guide part **224** by bending the large diameter part **223** of the skirt wall **220** so as to be inclined outwards in the direction toward the lower end (end) thereof, and molding the curling tight-contact part **225** configured to tightly surround the outside of the reinforcement frame **111** of the food container **100** by curling the upper end of the guide part **224** in the inward direction so as to have a semicircular round cross-sectional shape while the upper end of the guide part **224** is connected to the round-shaped stepped part **222**. The guide part **224** is provided to facilitate coupling with the upper portion of the food container **100**, and it is preferable to form the guide part **224** and mold the curling tight-contact part **225** simultaneously.

(23) Meanwhile, in the embodiment of the present invention, in the primary molding step (S**20**) and the secondary molding step (S**30**), each molding device (molds **400** and **500**) is preheated to the temperature of 100° C. to 140° C. (120° C. in the embodiment) through a heater. In this heated state, it is preferable to compress the lid molding paper **330** and the paper lid **200** made primarily while applying heat thereto within the range of 0.5 to 2 seconds (one second in the embodiment) so as to plastically deform the synthetic resin coating **320**. In this short-time heating compression process, the synthetic resin coating **320** treated on each of the upper and lower surfaces of the lid molding paper **330** is not damaged by heat, and when the molding of the paper lid **200** is completed, the synthetic resin coating **320** is hardened again and remains in the molded state (plastically deformed state). That is, when the molding of the paper lid **200** is completed and the synthetic resin coating **320** is cooled, the molded states of the cover plate **210**, the small diameter part **221**, the guide part **224**, and the curling tight-contact part **225** are maintained as they are.

(24) In more detail, as shown in FIG. **9**, in the paper lid **200** made by the manufacturing method according to the present invention, the skirt wall **220** includes the small diameter part **221**

configured to cause the cover plate **210** and the open upper portion of the food container **100** to be spaced apart from each other, the guide part **224** configured to expand in diameter while forming the round-shaped stepped part **222** so as to allow the lower end of the small diameter part **221** to contact the upper end of the reinforcement frame **111** of the food container **100**, the guide part **224** being bent to be inclined outwards in the direction toward the end thereof so as to guide coupling between the paper lid **200** and the upper portion of the food container **100**, and the curling tight-contact part **225** configured to tightly surround the outside of the reinforcement frame **111** of the food container **100** by curling the upper end of the guide part **224** in the inward direction so as to have a semicircular round cross-sectional shape while the upper end of the guide part **224** is connected to the stepped part **222**.
(25) Accordingly, as shown in FIGS. **10** and **11**, when the food container **100** is coved with the paper lid **200** by pressing the same from the top (refer to the direction of the arrow in FIG. **10**), the curling tight-contact part **225** provided in the middle of the skirt wall **220** may stably cover the open upper portion of the food container **100** by tightly surrounding the entire outside of the reinforcement frame **111** of the food container **100**. In addition, the guide part **224**, formed at the edge of the skirt wall **220** and molded to be inclined outwards, has a function of allowing the paper lid **200** to smoothly cover the food container **100** downwards from the upper portion of the food container **100**. According to this configuration, the paper lid **200** may also be used in the automation line to cover the food container **100** by easily pressing the same downwards from the upper portion of the food container **100**.
(26) Further, since the paper lid **200** for the food container **100** according to the present invention is manufactured while applying heat thereto at the temperature of 100° C. to 130° C. in the primary molding step (S**20**) and the secondary molding step (S**30**), the layer of the synthetic resin coating **320** is plastically deformed to be molded more easily and stably.
(27) Additionally, in the paper plate **200** for the food container **100** according to the present invention, bending strength is increased by molding the cover reinforcement part **211** concavely recessed downwards at the central portion of the cover plate **210**, thereby having an effect of further improving product reliability of the paper lid **200**.
Industrial Applicability
(28) The present invention may be widely used in the field of “a paper lid for a food container and a method of manufacturing the same”, configured to increase strength of the food container by curling the edge of the paper lid using an eco-friendly paper material and to be reliably applied to the food container.

Claims

1. A method of manufacturing a paper lid for a food container, the paper lid configured for covering an upper portion of the food container having a reinforcement frame formed at an uppermost portion of a wall surface and curled outwards so as to have a circular cross-sectional shape, the method comprising: a blanking step of cutting out a lid molding paper having a size capable of molding the paper lid from base paper having upper and lower surfaces, each of which is treated with a synthetic resin coating having a thickness of 10 to 35 microns, and forming wrinkles radially on an edge of the lid molding paper so as to facilitate molding of the paper lid; a primary molding step of supplying the lid molding paper made in the blanking step to a primary molding device, wherein the primary molding step comprises: compressing, while applying heat to the lid molding paper at temperature of 100° C. to 140° C., the lid molding paper within the primary molding device to mold the paper lid, such that the paper lid forms a cover plate configured to cover the open upper portion of the food container while being spaced apart from the open upper portion and a skirt wall bent downwards along an edge of the cover plate and configured to be coupled to the food container so as to surround an outside of the reinforcement frame of the food container, and

bending the skirt wall so as to form a small diameter part connected to the edge of the cover plate, the small diameter part having a lower end configured to contact with an upper end of the food container and a large diameter part formed to expand in diameter while forming a round-shaped stepped part configured to allow the lower end of the small part to contact an upper end of the reinforcement frame of the food container, the large diameter part configured to surround an outside of the upper end of the food container while being spaced apart from the outside thereof; and a secondary molding step of moving the paper lid made in the primary molding step to a secondary molding device, wherein the secondary molding step comprises: forming, while applying heat to the paper lid at temperature of 100° C. to 140° C., a guide part by bending the large diameter part of the skirt wall so as to be inclined outwards in a direction toward a lower end thereof, the guide part configured to guide a coupling with the upper portion of the food container, and molding a curling tight-contact part configured to surround the outside of the reinforcement frame of the food container by curling an upper end of the guide part an inward direction so as to have a semicircular round cross-sectional shape while the upper end thereof is connected to the round-shaped stepped part.

2. The method according to claim 1, wherein the synthetic resin coating is made of at least one of polyethylene (PE), polypropylene (PP), polylactic acid (PLA), polyethylene terephthalate (PET), or silicone.

3. The method according to claim 1, wherein the primary molding step further comprising molding a cover reinforcement part being concavely recessed downwards at a central portion of the cover plate.